HARSH NANAVATI

Applied AI Engineer | LLMs · RAG · Agentic Systems · Evaluation

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SUMMARY

Applied AI engineer with 2+ years shipping production LLM systems end to end: enterprise RAG, agentic pipelines with automated verification, real-time self-hosted LLM services and GPU computer vision, for a US client and in-house products. The largest serves **8,800+ people**, and others cut manual work by an estimated **40–70%**. I own the full path: retrieval design, guardrails, evaluation, FastAPI backends and deployment.

SKILLS

GenAI: LLMs, RAG, agentic workflows, LangGraph, LangChain, OpenAI API, Ollama (self-hosted LLMs), structured outputs, prompt engineering, guardrails, verification harnesses, human-in-the-loop

Retrieval and Evaluation: FAISS, vector search, OpenAI embeddings, Docling, chunking strategies, reranking, citation grounding, RAG evaluation (RAGAS)

ML and Computer Vision: PyTorch, scikit-learn, FaceNet/MTCNN, YOLOv8, ONNX, Elo and probabilistic models, backtesting

Deployment and MLOps: Docker, Docker Compose, Nginx, Prometheus and Grafana, CI/CD (GitHub Actions)

Engineering: Python, FastAPI, asyncio, REST APIs, SQL Server, PostgreSQL, MongoDB, Redis, Spark/Scala, Git, Angular

EXPERIENCE

SDE – Artificial Intelligence (Applied AI Engineer) | Masterly Solutions Pvt. Ltd. Oct 2024 – Present

Ahmedabad, India · Building in-house AI products and client systems for a US organization

- **DocuPilot – Enterprise RAG Platform:** cut procedure questions to supervisors by an estimated **~40%** for **600+ staff** across 14 document categories, with Docling ingestion (OCR and tables), custom structure-aware chunking with page-level provenance, and text-embedding-3-small with FAISS retrieval over **12,000+ chunks** at **sub-100 ms** median latency.
- **DocuPilot – Answer Grounding & Guardrails:** GPT-4o-mini answers constrained by answer-grounded citation selection and a two-stage citation gate, with every claim linked to its source page region; role-based document access, JWT auth, SHA-256 dedup and per-query observability (latency, citations, feedback) on FastAPI and SQL Server.
- **CrickPal Intelligence Layer – AI Win-Prediction Platform:** an LLM agent turns analysts' plain-English specs into feature code, gated by an automated verification harness (golden examples, symmetry, replay-based leakage, determinism) with self-repair and human approval; Elo baseline at **log-loss 0.643 and Brier 0.227** over 4.1k T20Is; **~800 automated tests**.
- **CrickPal AI Summary – Real-Time LLM Commentary Engine:** live ball-by-ball data turned into narrative and headline summaries on self-hosted Ollama; incremental summarization keeps prompts flat as a match grows; exactly-once batching on one instance, a circuit breaker with backoff and per-match prompt and model tuning without redeploys.
- **AI Face Matching System – GPU Photo-Tagging Pipeline:** cut manual photo tagging by an estimated **~70%** with an unattended MTCNN and FaceNet pipeline: self-healing job queue, retry budgets, stale-lease recovery and an embedding cache across three SQL Server databases; extended to video with YOLOv8 detection and ONNX/DirectML export.
- **QuantumCopilot – AI Loan & Financial Assistant:** maps natural-language queries to typed parameters across 6 loan products; parallel scenario fan-out (an estimated **~3x faster**), multi-turn memory, and 8 Uvicorn workers on IIS.
- **RAG Support Chatbot – Customer Support Assistant:** handled **30,000+ questions for 8,800+ people** in production since May 2025, ~86% without a fallback reply; FAISS retrieval over question-and-answer chunks and a staff correction loop that writes fixes back into the knowledge base (**~60% fewer escalations**, estimated).
- **Concierge Chatbot – Long-Context Visitor Assistant:** redesigned from a RAG pipeline to ~50 FAQs in one cached prompt, with single-call intent routing and fixed replies for out-of-scope requests.

Intern – Data Science & Big Data | Deloitte Haskins & Sells LLP, Remote, Ahmedabad, India Jun 2023 – May 2024

- Used Spark, Scala and SQL for entity-resolution and anomaly-detection analyses on large-scale financial datasets.
- Built Power BI dashboards for enterprise audit reporting workflows.

PROJECT & RESEARCH

ViperMind – Multi-Agent Python Tutor (personal project): a LangGraph router over four GPT-4 agents; code grades answers against an LLM-written key; Redis cache, PostgreSQL and a 7-service Docker Compose stack with monitoring.

Blockchain-Driven Federated Learning for Parkinson's Detection (M.Tech research): privacy-preserving, explainable AI.

EDUCATION

M.Tech, Data Science | Nirma University, Ahmedabad | CGPA 8.79/10 2022 – 2024

B.E., Computer Engineering | LDRP-ITR, Gandhinagar | CGPA 8.75/10 2017 – 2021